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Week 5 Day 4

KPI (Key Performance Indicator)

It is a measurable value that is used to evaluate how effective, efficiency and quality of the software testing process.

Testing Metrics

It is any measurable value used to track or understand the performance of testing.

KPI

It is a specific and important metric used to measure whether a particular testing or business goal is being achieved.

1. Defect Metrics

a) Defect Density

It measures how many defects exist compared with the size of the software.

$$\text{Defect Density} = \frac{\text{No. of defects}}{\text{No. of KLOC}}$$

b. Defect Arrival Rate

It measure how quickly new defects are being reported.

$$\text{Arrival Rate} = \frac{\text{No. of defects}}{\text{No. of days}}$$

c. Defect Severity Index (DSI)

It measure the overall seriousness of defects found in the application.

$$\text{DSI} = \frac{(\sum (\text{Defects} \times \text{Severity Weight}))}{\text{Total no. of defects.}}$$

d. Defect Removal Efficiency (DRE)

It measure how many defects are found before the software reaches customers.

$$\text{DRE} = \frac{\text{Defects Removed}}{(\text{Defects Removed} + \text{Escaped Defects})} \times 100$$

e. Defect Leakage

It measure the percentage of defects that escape testing and reach production.

$$\text{Defect Leakage} = \frac{\text{Defects found in Production}}{\text{Total Defect}} \times 100$$

f. Reopen Rate

It measure how often fixed defects are reopend because ~~that~~ the problem was not properly fixed.

$$\text{Reopen Rate} = \frac{\text{Reopened Defects}}{\text{Total Fixed Defects}} \times 100$$

2. Test Execution Metrics

a. Test Case Execution Rate

It measure how many planned test cases have actually being executed.

$$\text{Execution Rate} = \frac{\text{Executed Test Cases}}{\text{Planned test cases}} \times 100$$

b. Test Pass / fail Percentage

It measure how many executed tests pass or fail.

$$\text{Pass Rate} = \frac{\text{Passed Tests}}{\text{Total no. of test}} \times 100$$

c. Test Case Effectiveness

It measure how well test cases discover actual defects.

$$TCE = \frac{\text{Defects Detected}}{\text{Test Cases Run}} \times 100$$

d. Test Case Productivity

It measure how many test cases a tester creates within a certain time.

$$\text{Productivity} = \frac{\text{Number of Test Case Designed}}{\text{Effort}}$$

3. Coverage Metrics.

a. Requirement Coverage

It measure how many software requirements have corresponding tests.

$$\text{Test Coverage} = \frac{\text{Requirements Covered}}{\text{Total Requirements}} \times 100$$

b. Code / Statement Coverage

It measures how much of the application code is executed by automated tests.

$$\text{Code Coverage} = \frac{\text{Line of Code Executed}}{\text{Total Line of Code}} \times 100$$

c. Risk Coverage

It measures whether the most important / high-risk parts of the application are tested.

$$\text{Risk Coverage} = \frac{\text{No. of test}}{\text{No. of Risk area}} \times 100$$

4. Effort & Cost Metrics

a. Test Effort Variance

It measures the difference between planned testing time and actual testing time.

$$\text{Effort Variance} = \text{Estimated time} - \text{Actually time}$$

b. Cost per Defect

It measures the average cost spent to discover one defect.

$$\text{Cost per Defect} = \frac{\text{Total Testing Cost}}{\text{Total No. of Defect found}}$$

c. Testing ROI (Return on Investment)

It measures the financial return gained from testing compared with the testing investment.

$$\text{ROI} = \frac{\text{Manual Testing Cost} - \text{Automatic Testing Cost}}{\text{Automatic Testing Cost}} \times 100$$

5. Automation Metrics.

a. Percentage of Automated Tests.

It measures how many test cases are automated instead of manually executed.

$$\text{Automation Coverage} = \frac{\text{Number of Auto. Test Case} \times 100}{\text{Total Test Case}}$$

b. Test Script Maintenance Effort

It measures how much time testers spend maintaining automated test scripts.

$$\text{Maintenance Effort} = \frac{\text{Hours spent Updating tests}}{\text{Total Test Automation Hrs.}}$$

c. Flaky Test Rate

It measures how many automated tests fail randomly even though the application is working correctly.

$$\text{Flaky test Rate} = \frac{\text{No. of Flaky Test}}{\text{Total Automated test}} \times 100$$

d. Automation ROI

It measures how much money automation saves compared with its cost.

$$\text{ROI} = \frac{(\text{Manual cost} - \text{Automated Cost}) \times 100}{\text{Automated Testing Cost}}$$

• Python Logging

The logging module in Python is used to record information about what is happening inside a program.

Log Level

- ~~Debug~~ Debug - It is used for detailed information mainly during development and debugging - for eg:

`logging.debug("User ID is 101")`

- Info - It is used to record normal application activities. for eg.

`logging.info("User logged in succ")`

- Warning - It indicate something unexpected or potentially problematic, but the application can continue - for eg.

`logging.warning("Password will expire soon")`

- Error - It indicate that an operation is failed. for eg:

`logging.error("Payment failed")`



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• Critical - It indicates a very serious problem that may cause the application to stop. for eg:-

logging.critical ("Database server is down")

Log files

It is a file where application log messages are stored instead of displaying them only on the console. for eg.

logging.basicConfig (

filename="SNL.log";

level=logging.INFO

)